

Smart Slot Booking

Faculty office hours & lab slot scheduling

A clean-room rebuild focused on timezone correctness, buffers, and concurrency.

THE PROBLEM

Students and faculty can face timezone confusion, buffer violations, and double-booking under simultaneous requests.

THE SOLUTION

UTC-backed scheduling, host-local recurrence rules, database-level exclusion, atomic reservation holds, and explainable recommendations.

RESULT

11 automated tests passing, including all 3 mandatory Killer Tests and focused tests for the differentiator.

Architecture & Core Scheduling

1 AVAILABILITY

Host recurring rules and date overrides are evaluated in the host's IANA timezone. Candidate slots are converted to UTC instants.



2 CONFLICT ENGINE

Existing bookings are expanded with pre/post buffers. Candidate ranges that overlap protected intervals are rejected.



3 BOOKING

A short SelectedSlots hold protects checkout. Confirmation rechecks availability and relies on PostgreSQL exclusion for the final race.

The 3 Mandatory Killer Tests

KT1 TIMEZONE

Host: Asia/Kolkata

Booker: America/Los_Angeles

10:00–11:00 IST becomes 21:30–22:30 PDT on the previous calendar day.

KT2 BUFFERS

15-minute pre/post buffers protect an existing 14:30–15:00 booking. Only the valid non-overlapping slot remains available.

KT3 CONCURRENCY

Two simultaneous requests for the same slot enter transactions. Exactly one commits; the other receives 409 SLOT_ALREADY_BOOKED.

✓ 3/3 Killer Tests PASS

GAPS Fix + Differentiator

GAPS IMPROVEMENT

PostgreSQL GiST exclusion constraint with btree_gist prevents overlapping buffered bookings for the same host at the database layer.

Atomic SelectedSlots reservation holds are also validated and consumed inside the booking transaction.

SMART SLOT RECOMMENDATION

Students can provide a preferred booking time. Available slots are ranked by distance from that preference and the best 3 are highlighted.

No AI API or external service is required; ranking is deterministic.

Explainable Scheduling

WHY THIS SLOT?

Recommended slots explain their ranking: best match, Pick #2, Pick #3, or earliest available when no preference is supplied.

TIMEZONE CLARITY

The UI can show both host time and booker local time, making cross-calendar-day conversions understandable to students.

UNAVAILABILITY REASONS

Buffer conflicts and host day-off overrides receive user-friendly explanations instead of opaque failures.

11/11 automated tests passing

Clean-room implementation • No Cal.com source reuse • No Redis • No AI dependency